

STAT 5350

Applied Deep Learning 1

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	AI Engineer (74.8 / 100)	65.8 / 100

66

CURRENT AVG SCORE
across 6 job roles

97

POTENTIAL SCORE
with recommended additions

+39

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces students to state-of-the-art deep learning techniques employed in the industry. This course will focus on training neural networks and computer vision, including image classification and transformation, object detection, and facial recognition. Advanced topics will include domain adaptation and learning techniques. There will be an emphasis on reading current literature.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	74.8	100.0	+25.2	MEDIUM
Applied Scientist	71.7	100.0	+28.3	MEDIUM
Machine Learning Engineer	71.6	100.0	+28.4	MEDIUM
ML Engineer	68.9	100.0	+31.1	HIGH
Data Scientist	58.9	93.5	+34.6	HIGH
MLOps Engineer	48.7	87.2	+38.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

Deep Learning 1 of 6 roles

cnn

Experiment Tracking 1 of 6 roles

tracking

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 HIGH Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 MEDIUM Consider Business Communication content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

3 LOW Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: cnn.

4 LOW Optional: Experiment Tracking — would broaden appeal to one additional role. Candidate terms: tracking.

GAP KEYWORDS BY ROLE

AI Engineer

74.8 → 100.0 (+25.2 pts)

data models build technical systems problems technology machine

Applied Scientist

71.7 → 100.0 (+28.3 pts)

discovery scientific research problems models technical data systems

Machine Learning Engineer

71.6 → 100.0 (+28.4 pts)

data analytics business mobile design science models applications

ML Engineer

68.9 → 100.0 (+31.1 pts)

cnn discovery data machine systems models world build

Data Scientist

58.9 → 93.5 (+34.6 pts)

data analytics presentation insights different social build problems

MLOps Engineer

48.7 → 87.2 (+38.6 pts)

data text systems machine tracking models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.